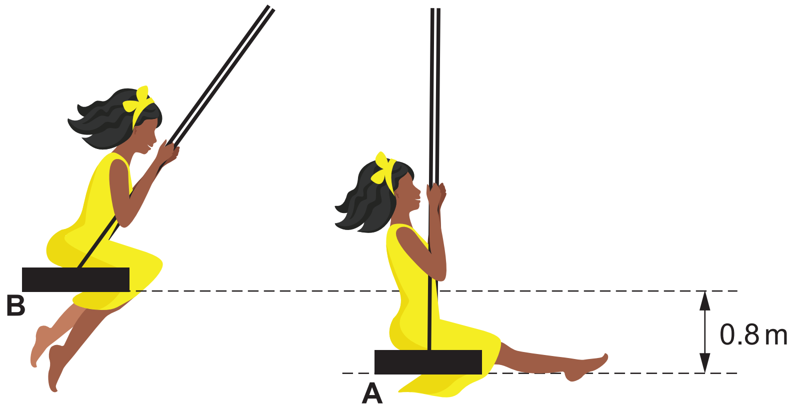
Examiner only

4. A child of mass 35 kg is playing on a swing.



The swing is pulled back to position **B**, 0.8 m above the start point, and released.

(a) (i) Use an equation from page 2 to calculate the potential energy gained by the child when moving from position **A** to position **B**. [2]
[Gravitational field strength, g = 10 N/kg]

potential energy = ... J

(ii) State the kinetic energy of the child as they pass through point **A**.
Assume there are no resistive forces acting. [1]

kinetic energy = ... J

(iii) Use an equation from page 2 to calculate the velocity of the child as they pass through point **A**. [3]

velocity = ... m/s

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(b) (i) At position **B** the child drops a ball which hits the ground time $t = 0.50\,\text{s}$ later.
The ball has an initial velocity, $u = 0\,\text{m/s}$.
The downwards acceleration, $a = g = 10\,\text{m/s}^2$.

Use an equation from page 2 to determine the distance, x, that the ball falls. [3]

x = ... m

(ii) After the collision with the ground, the ball rolls away with initial kinetic energy of 6.4 J and then comes to a halt in 3.0 m.

Use an equation from page 2 to calculate the size of the friction force between the ball and the ground. [2]

friction force = ... N

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